Math 443 Bonus HW 2

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Bonus Problem 1

Algorithm 1 Gaussian Elimination - Regular Case

```
start
for j = 1 to n do
    if M_jj = 0 then
        stop; print "A is not regular"
    else
        for i = j + 1 to n do
            set l_ij = m_ij / m_jj
            add -l_ij times row j of M to row i of M
        end for
    end if
end for
```

Compare the cost of the Inverse matrix approach and Gaussian elimination. Which one is more efficient?

Proof. **Gaussian Elimination:**

Initially, we can see we do a single arithmetic check for if $M_jj = 0$ for all n, this happens each time and we will add it to our final computation. Furthermore, we will be considering multiplication and addition as being equivalent to division and subtraction. We then start the first for loop by performing $n-1$ operations of $l_{ij} = m_{ij}/m_{jj}$ ($n-j$ generally). Each of these operations will need to be multiplied by the jth row, in the first iteration of the loop this is $n+1$ (assuming an augmented matrix of $Ax = b$) (or $n-j+2$ generally) multiplication operations for *each* iteration of the outer for loop, which then gets added to each row. For each row this is a subtraction (or addition) of $n+1$ operations (this can arguably be n operations because you know the jth column will always be 0, but this algorithm does not consider that), making $n-j+2$ operations generally. This gives us

$$n \text{ arithmetic checks}$$

$$\sum_{j=1}^{n}(n-j)(n-j+2) \text{ multiplication operations}$$

$$\sum_{j=1}^{n}(n-j)(n-j+2) \text{ addition operations}$$

Which evaluates to to

$$\sum_{j=1}^{n}(n-j)(n-j+2) = \frac{1}{6}(2n^3+3n^2+n)$$

This article was used for simplifying summations. Furthermore, we need to perform back substitution for the final answer, which according to the textbook is $\frac{n^2-n}{n}$ addition and multiplication operations, which yields a total:

$$\frac{1}{3}n^3+\frac{1}{2}n^2+\frac{7}{6}n-1 \text{ multiplication operations}$$

$$\frac{1}{3}n^3+\frac{1}{2}n^2+\frac{7}{6}n-1 \text{ addition operations}$$

$$n \text{ arithmetic checks}$$

Matrix Inverse Approach:

Initially, we must calculate the inverse of a matrix through Gaussian elimination. Naively, we can claim that we need $2n$ multiplication and addition operations for the row adding step of the algorithm. But, we know that because the augmented matrix is made with the identity matrix, the multiplication factors are 1 and 0, and we don't need to compute the 0 multiplication, nor addition steps. Thus, we can simplify the algorithm to for the first iteration $n+1$ operations. The next step will require $n+2$, thus we get $(n+j)$ operations generally. Thus, we get the following:

$$\frac{4}{3}n^3+\frac{1}{2}n^2+\frac{1}{6}n \text{ multiplication operations}$$

$$\frac{4}{3}n^3+\frac{1}{2}n^2+\frac{1}{6}n \text{ addition operations}$$

Afterwards, we just perform matrix vector multiplication, which according to the slides give us n^2 multiplication operations and $n^2 - n$ addition operations, which finally yields

$$\frac{4}{3}n^3 + \frac{3}{2}n^2 + \frac{1}{6}n \text{ multiplication operations}$$
$$\frac{4}{3}n^3 + \frac{1}{2}n^2 - \frac{5}{6}n \text{ addition operations}$$

Which is clearly significantly slower than Gaussian Elimination, insofar as the inverse is not given or abundantly clear.

$\square$